

2. Object Detection Extension

Problem Statement

Different object detection use cases require different trade-offs between **accuracy, speed, and computational cost**. A single model cannot satisfy all scenarios.

Approach

Developed a **unified object detection extension** using **PyTorch**, supporting multiple architectures:

- Faster R-CNN
- FCOS
- SSD
- SSD Lite
- RetinaNet
- YOLO (extended work)

Training

- Dataset annotated using **XML bounding boxes**
- Transfer learning using **COCO pre-trained weights**
- Fully configurable training pipeline:
 - Model selection
 - Batch size
 - Epochs
 - Learning rate
 - Momentum
 - Weight decay
- Evaluation metrics:
 - IoU (Intersection over Union)
 - Precision
 - Recall (per class)

Inference

- Users can choose any trained or pre-trained model
- Output includes:
 - Image with bounding boxes, labels, and confidence scores
 - Tabular output with detected objects
- Results also generated in **base64 format** for easy deployment and integration

Key Observations

- Faster R-CNN performed best for accuracy
- SSD and SSD Lite were faster but had lower recall
- RetinaNet handled class imbalance well
- Model choice depends on application requirements